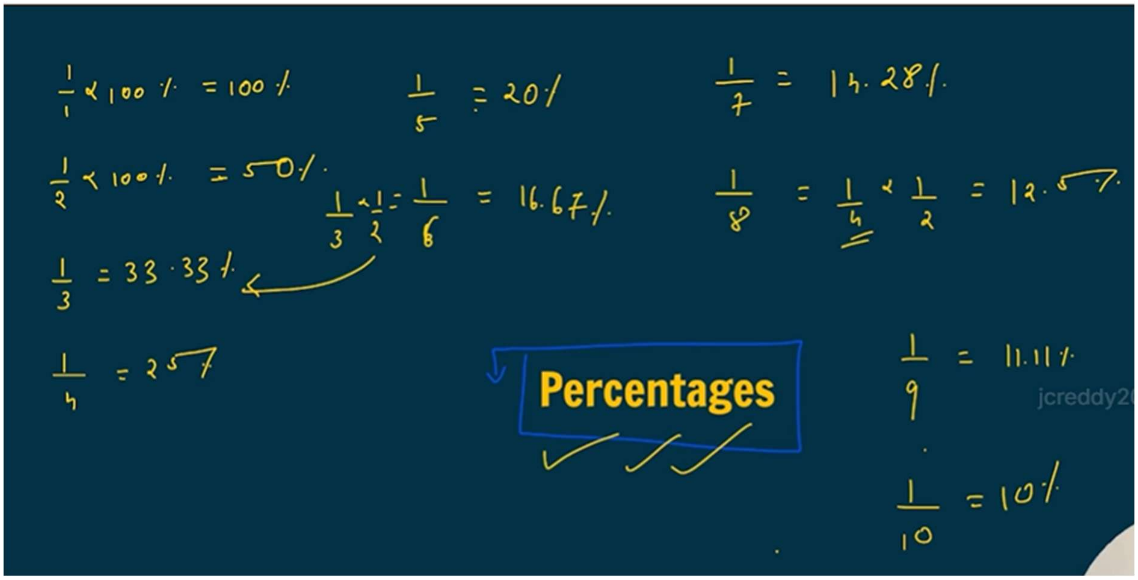
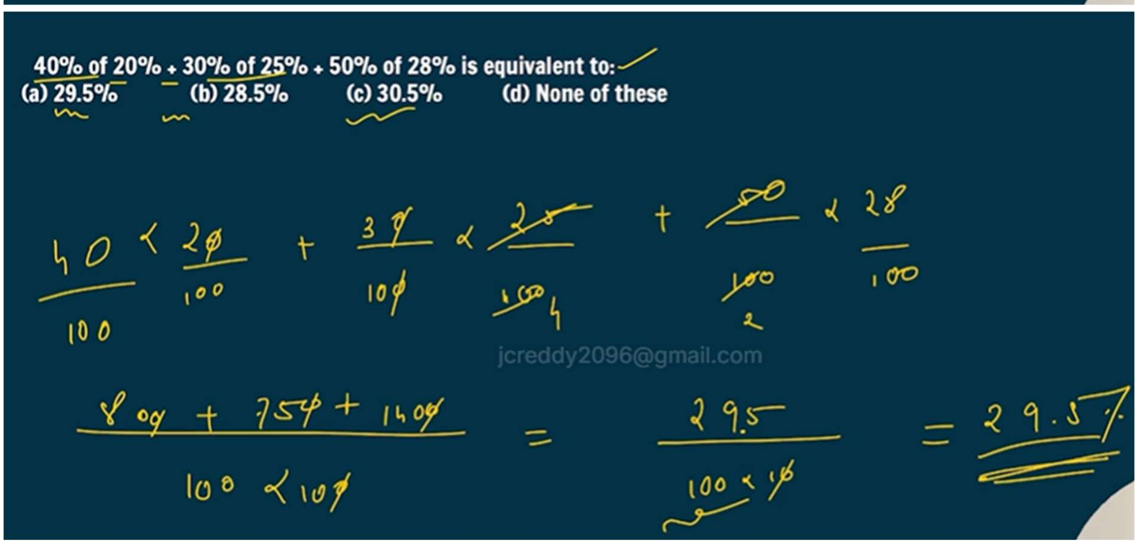
→Percentages





If a number is 20% more than the other, how much per cent is the second number less than the first?
(a) 12.33% (b) 16.66% (c) 16.33% (d) None of these

① ②

$100 \xrightarrow{20\%} 120$

f i

$\boxed{\frac{f - i}{i} \times 100} \rightarrow$ change

$\frac{100 - 120}{120} \times 100$

$= -\frac{\cancel{20}^{1}}{\cancel{120}_{6}} \times 100$

$= -16.67\%$

If the given two numbers are respectively 7% and 28% of a third number, then what percentage is the first of the second?
(a) 20% (b) 25% (c) 18% (d) None of these

1 2 3

28 7 100

28% 7%

$\frac{2}{1} \times 100$

$\frac{\cancel{7}}{\cancel{28}_{4}} \times 100$

$= 25\%$

The price for a pair of cuff links is Rs 1.00. The price for a 5-pair package of cuff links is Rs 3.40. The 5-pair package is what percent cheaper per pair than 5 pairs purchased separately?
(a) 32% (b) 47% (c) 62% (d) 63%

1 Pair = 1 Re → 5 Pair = 5 Rs
i

5 Pair = 3.4 Re
f

$\frac{1.6}{\cancel{5}} \overset{20}{\cancel{\times 100}} = 32\%$

In two successive years 100 and 75 students of a school appeared at the final examination. Respectively 75% and 60% of them passed. The average rate of pass is

(a) $68\frac{4}{7}$% (b) 78% (c) 80% (d) $80\frac{1}{2}$%.

$100 \xrightarrow{0.75} 75$

$75 \xrightarrow{\times 0.6} 45$

$175 \qquad 120$

$\frac{120}{175} \times 100 = 68....$

Rajan got 76 percent marks and Sonia got 480 marks in a test. The maximum marks of the test is equal to the marks obtained by Rajan and Sonia together. How many marks did Rajan score in the test?

(a) 1450 (b) 1520 (c) 1540 (d) 2000 (e) None of these

$R = 76\%$

$S = 480$

$R + S = 100\%$

$76\% + 480 = 100\%$

$480 = 24\%$

$\boxed{2000 = 100\%}$

76%

1520

Vinay decided to donate 5% of his salary. On the day of donation he changed his mind and donated Rs 1687.50, which was 75% of what he had decided earlier. How much is Vinay's salary?
(a) Rs 33750 (b) Rs 37500 (c) Rs 45000 (d) Cannot be determined (e) None of these

$$\frac{\cancel{75}^{15}}{100} \times \frac{\cancel{5}}{\cancel{100}\ 20_4} \times x = 1687.50$$

intially

$$x = \frac{168750}{15} = 45000$$

→DAY-2

In a competitive examination in State A, 6% candidates got selected from the total appeared candidates. State B had an equal number candidates appeared and 7% candidates got selected with 80 more candidates got selected than A. What was the number of candidates appeared from each State?
(a) 7600 (b) 8000 (c) 8400 (d) Data inadequate

$7\% - 6\% = 80$

$1\% = 80$

$100\% = 8000$

Shortcut

A monthly return railway ticket costs 25 percent more than a single ticket. A week's extension can be had for the former by paying 5 percent of the monthly ticket's cost. If the money paid for the monthly ticket (with extension) is Rs 84, the price of the single ticket is

(a) Rs 48 (b) Rs 64 (c) Rs 72 (d) Rs 80

single $= x$

Return $= \frac{5x}{4} =$

Ext $= \frac{5x}{4} \times \frac{\cancel{105}^{21}}{\cancel{100}_{\cancel{20}\,4}} = \frac{\cancel{21}x}{16} = \cancel{84}^{4}$

$x = 64$

10% of the voters did not cast their vote in an election between two candidates. 10% of the votes polled were found invalid. The successful candidate got 54% of the valid votes and won by a majority of 1620 votes. The number of voters enrolled on the voters' list was

(a) 25000 (b) 33000 (c) 35000 (d) 40000

54, 46% $\} = 8\%$

100%

Casted $= 90\%$

10% ↓

Valid $= 81\%$

win $= \frac{8}{100} \times \underbrace{\frac{81}{100}}_{\text{valid}} \times x = 1620$

$\boxed{x = 25{,}000}$

A company received two shipments of ball bearings. In the first shipment, 1 percent of the ball bearings were defective. In the second shipment, which was twice as large as the first, 4.5 percent of the ball bearings were defective. If the company received a total of 100 defective ball bearings, how many ball bearings were in the first shipment?

(a) 990 (b) 1000 (c) 2000 (d) 3000

$$\frac{1}{100} \times x + \frac{4.5}{100} \times 2x = 100$$

$$\frac{10x}{100} = 100$$

$$\boxed{x = 1000}$$

A businessman imported Laptops, worth Rs 210000, Mobile phones worth Rs 100000 and Television sets worth Rs 150000. He had to pay 10% duty on laptops, 8% on phones and 5% on Television sets as a special case. How much total duty (in Rupees) he had to pay on all items as per above details?

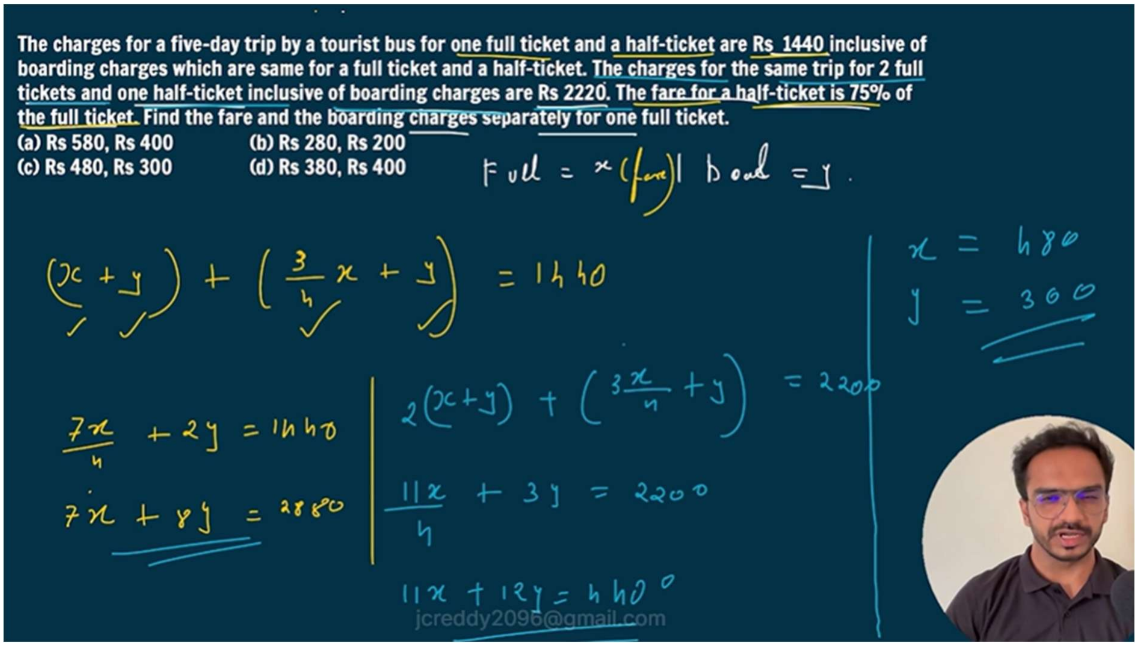
(a) 36500 (b) 37000 (c) 37250 (d) 37500

21000

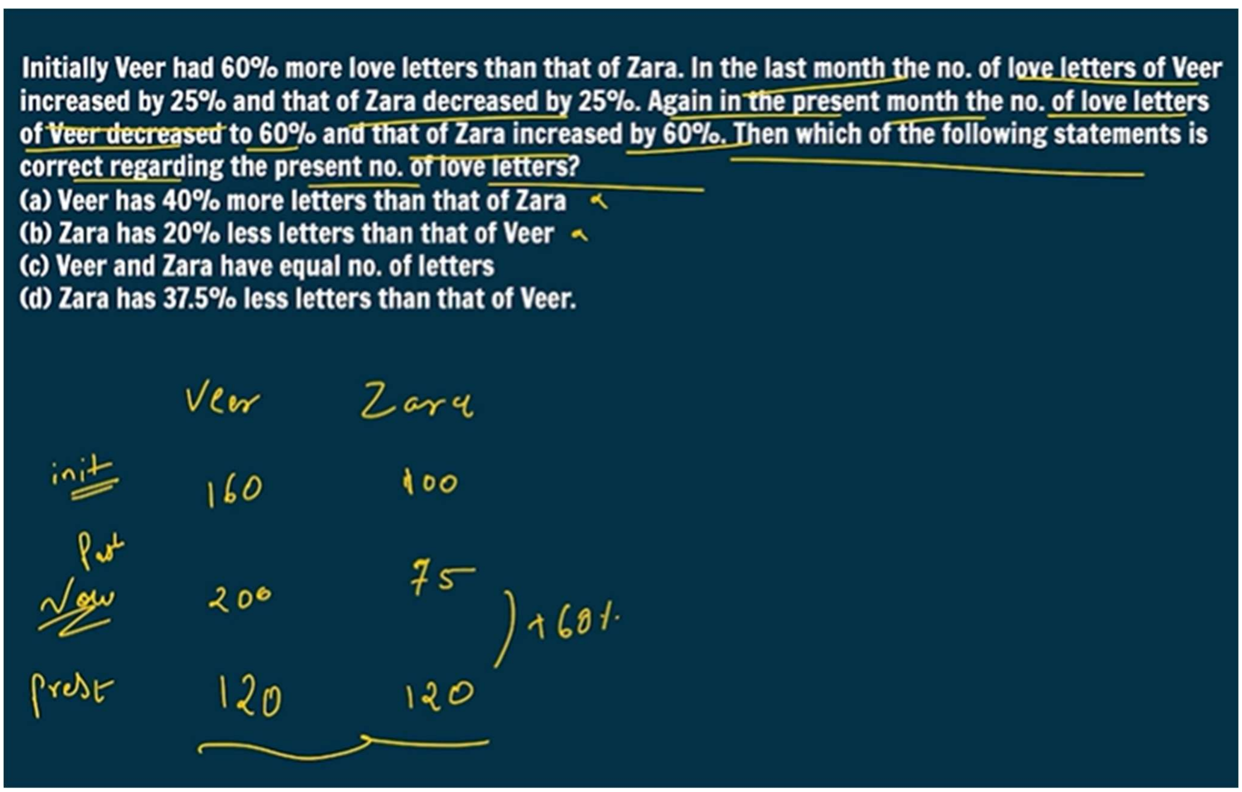
8000

7500

36500


The charges for a five-day trip by a tourist bus for one full ticket and a half-ticket are Rs 1440 inclusive of boarding charges which are same for a full ticket and a half-ticket. The charges for the same trip for 2 full tickets and one half-ticket inclusive of boarding charges are Rs 2220. The fare for a half-ticket is 75% of the full ticket. Find the fare and the boarding charges separately for one full ticket.
(a) Rs 580, Rs 400
(b) Rs 280, Rs 200
(c) Rs 480, Rs 300
(d) Rs 380, Rs 400
Full = x (fare) | board = y.
$(x + y) + \left(\frac{3}{4}x + y\right) = 1440$
$\frac{7x}{4} + 2y = 1440$
$7x + 8y = 2880$
$2(x+y) + \left(\frac{3x}{4} + y\right) = 2200$
$\frac{11x}{4} + 3y = 2200$
$11x + 12y = 4400$
$x = 480$
$y = 300$
jcreddy2096@gmail.com

→DAY-3


Initially Veer had 60% more love letters than that of Zara. In the last month the no. of love letters of Veer increased by 25% and that of Zara decreased by 25%. Again in the present month the no. of love letters of Veer decreased to 60% and that of Zara increased by 60%. Then which of the following statements is correct regarding the present no. of love letters?
(a) Veer has 40% more letters than that of Zara
(b) Zara has 20% less letters than that of Veer
(c) Veer and Zara have equal no. of letters
(d) Zara has 37.5% less letters than that of Veer.
Veer Zara
init 160 100
Past Now 200 75
+60%
prest 120 120

My friend Siddhartha Ghosh is working in the Life Insurance Corporation of India (LIC). He was hired on the basis of commission and he got the bonus only on the first years commission. He got the policies of Rs 2 lakh having maturity period of 10 year. His commission in the first second, third, fourth and for the rest of the years is 20%, 16% 12% 10% and 4% respectively. The bonus is 25% of the commission. If the annual premium is Rs 20,000 then what is his total commission if the completion of the maturity of all the policies is mandatory?

(a) Rs 17,400 (b) Rs 23, 600
(c) Rs 15,000 (d) Rs 15,500

10

20, 16, 12, 10   6 = 4%

1st 20% of 20,000 = 4000 ✓

Bonus = 1000 ✓

2nd = 3200 ✓

3rd = 2400 ✓

4th = 2000 ✓

= 800 × 6 = 4800 ✓ 17400

In the Presidency College two candidates contested a presidential election. 15% of the voters did not vote and 41 votes were invalid. The elected contestant got 314 votes more than the other candidate. If the elected candidate got 45% of the total eligible votes, which is equal to the no. of all the students of the college. The individual votes of each candidate are :

(a) 2250 and 1936 (b) 3568 and 3254
(c) 2442 and 2128 (d) 2457 and 2143

vote = 85%

valid = 85% − 41 ✓

win − losser = 314

win = 314 + L

Total = x

Losser = y

2457   2143

$0.85x - 41 = 2y + 314$ — (1)

$2(0.45x = y + 314)$

$0.05x + 41 = 314$ | $x = 5460 \times 0.85$

$= 4641 - 41$

$= 4600$

In the Polo hospital some patients who were suffering from the Hepatites-B were admitted for the treatment , but 9% of the patients were died within half an hour. After treatment, the percentage of patients cured out of the remaining was only 80%. Out of these patients only 70% are completely cured out and the remaining were partially cured out which were equal to 153 patients. The no. of patients (approx.) who were admitted for the treatment for the same was :

(a) 400 (b) 678 (c) 560 (d) 700

100% = 700

91% = $\boxed{637.5}$ ≈ 637 = 91%

80%

$\boxed{570}$

70% 30% Part

357 153

The annual earning of Mr. Sikkawala is Rs 4 lakhs per annum for the first year of his job and his expenditure was 50%. Later on for the next 3 years his average income increases by Rs 40,000 per annum and the saving was 40%, 30% and 20% of the income. What is the percentage of his total savings over the total expenditure if there is no any interest is applied on the savings for these four years?

(a) $49\frac{37}{87}$% (b) $41\frac{73}{83}$% (c) 53% (d) none of the above

Income = 4 4.4 4.8 5.2

Saving = 2 1.76 1.44 1.04 = 6.24

exp = 2 2.64 3.36 4.16 = 12.16

$\frac{\text{Sav}}{\text{exp}} \times 100 = \frac{6.24}{12.16} \times 100$

= 51%

A company has 12 machines of equal efficiency in its factory. The annual manufacturing expenses are Rs 24,000 and the establishment charges are Rs 10,000. The annual output of the company is Rs 48,000. The annual output and manufacturing costs are directly proportional to the no. of machines while the share holders get the 10%profit,which is directly proportional to the annual output of the company. If 8.33% machines remained close throughout the year. Then the percentage decrease in the amount of Share holders is

(a) 16.66% (b) 14.28% (c) 8.33% (d) none of these

8.33 ≈ 1